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T13
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12/01/26 Assignment-01

Q. Software process models.

→ A software process model is a structured approach that defines the sequence of activities required to develop a software system. It provides a framework that specifies what tasks are to be performed, when they are performed, and how they are executed during the software development life cycle. (SDLC).

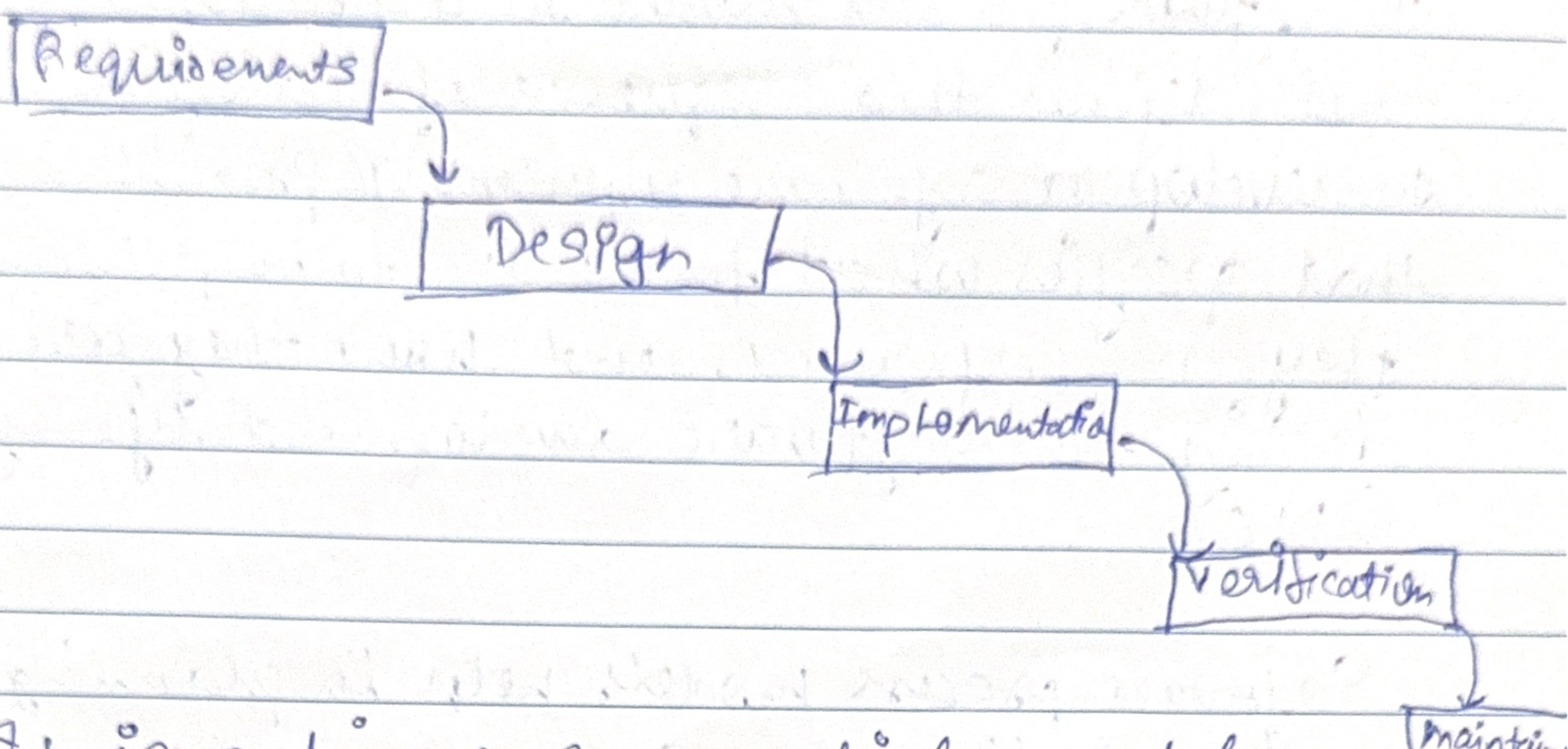
Software process models help in planning, organizing and controlling the software development process to ensure high-quality software is delivered within time and cost constraints.

PHASES

1. Requirement Analysis :- Understanding and documenting user requirements.
2. System design :- Designing software architecture & components.
3. Implementation :- Coding the software as per design.
4. Testing :- Verifying & validating the software
5. Deployment :- Delivering the software to the user.
6. Maintenance :- modifying & improving the software after delivery

Types of Software Process Models

1. Waterfall models :-

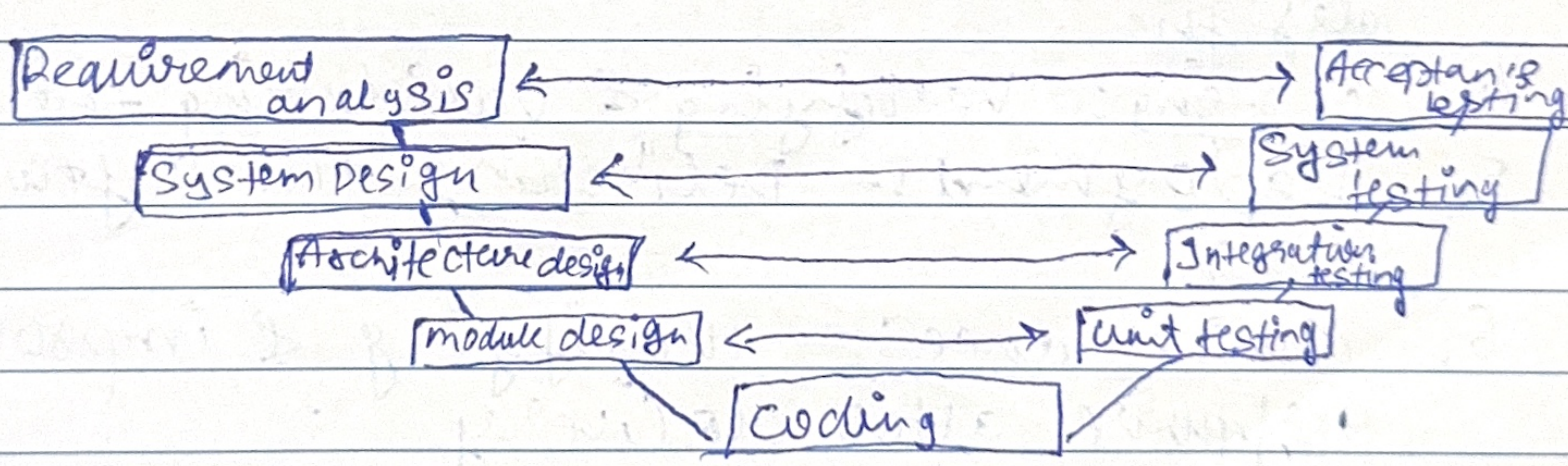


It is a lineas & sequential model where each phase must be completed before moving to the neset phase.

ADV
- Simple & easy to understand
- Suitable when requirements are well defined

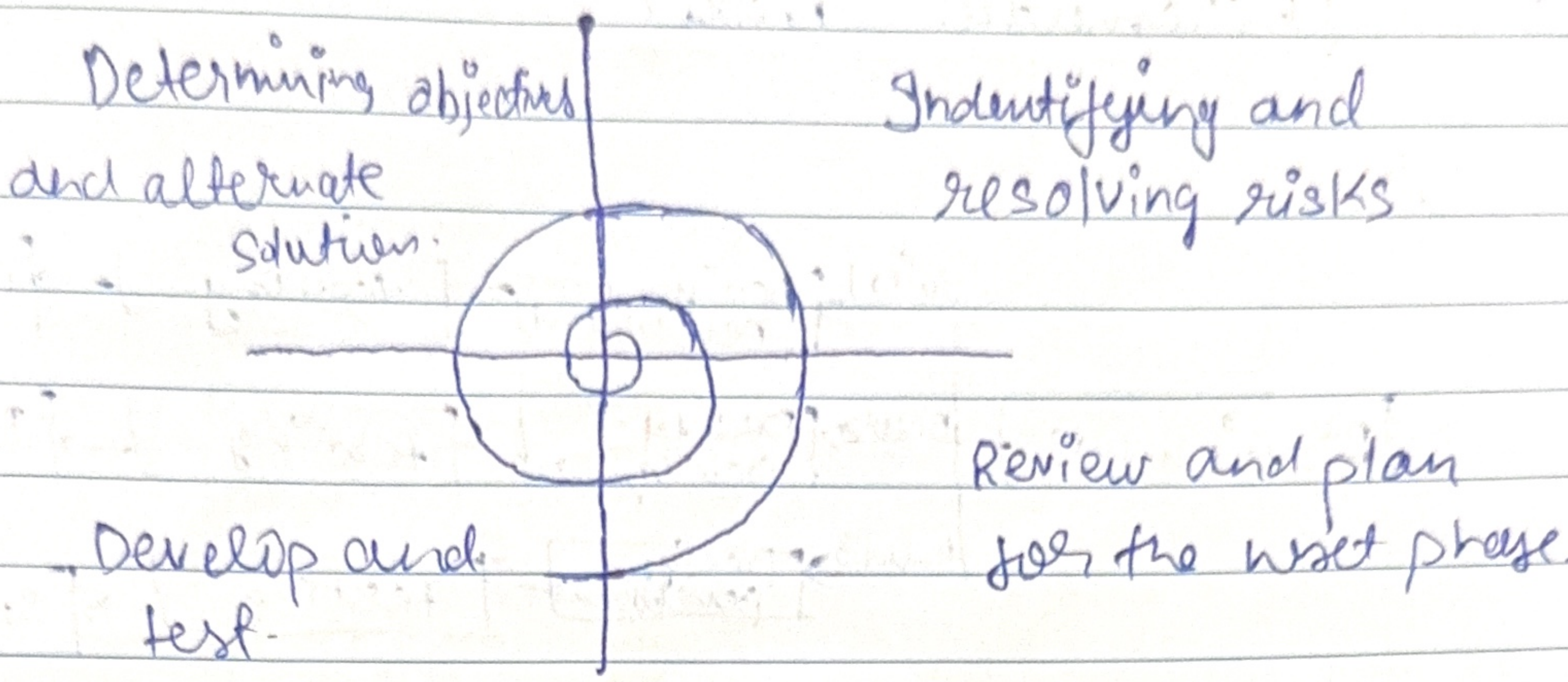
DIS-ADV
- Changes are difficult to implement.
- Errors are detected late.

2. V-model



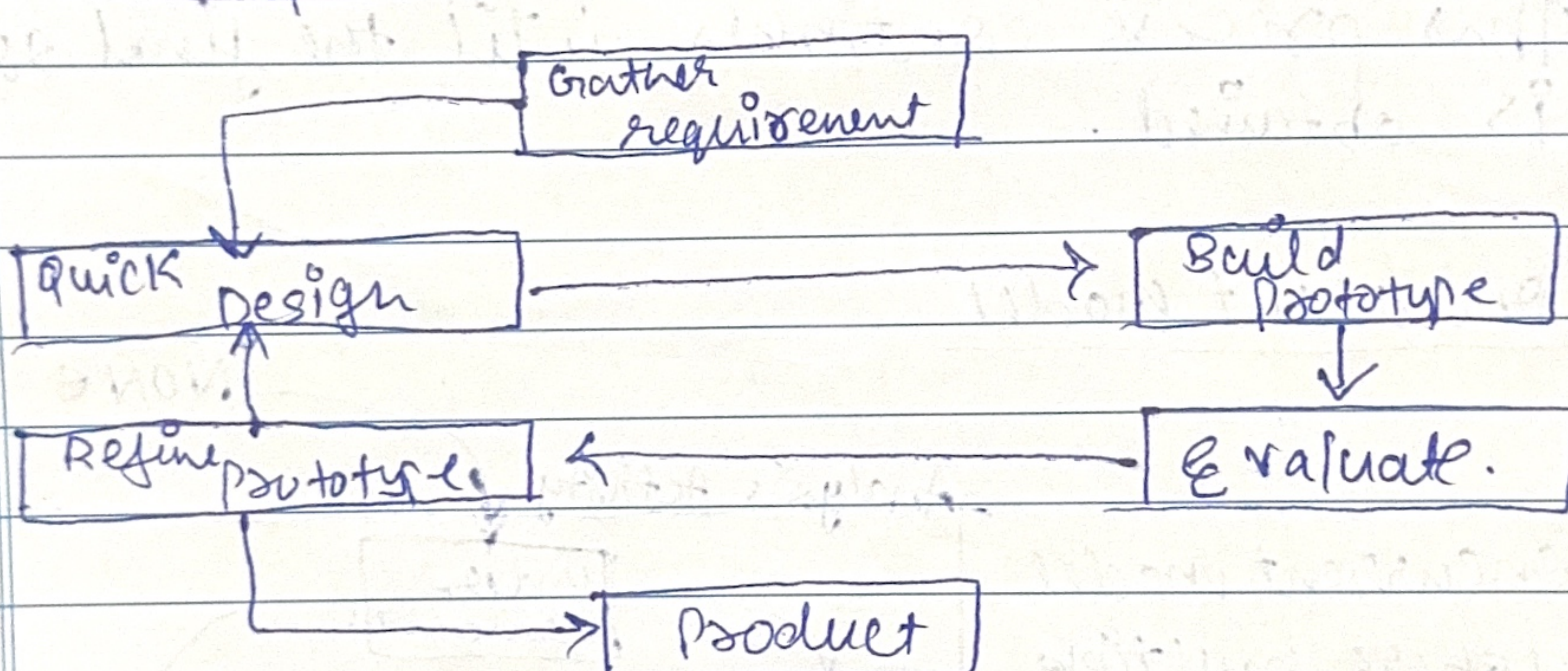
- V-model, also known as the Verification & validation model, is an extension of waterfall

Spiral Model



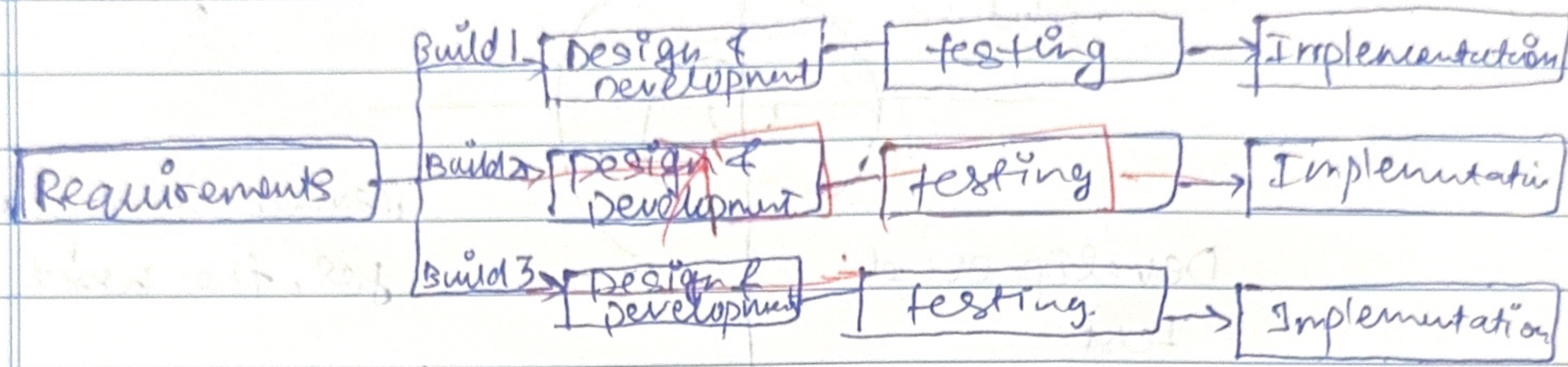
- Spiral Model is a software development process model. This model has characteristics of both iterative & waterfall models.
- Risk Handling feature, which identified in each phase & they are resolved through prototyping

4. ## Prototype Model



- Prototype model is an activity in which prototypes of software applications are created. First prototype is created & then the final product is manufactured based on that prototype.
- This model is created when we do not know the requirements well.

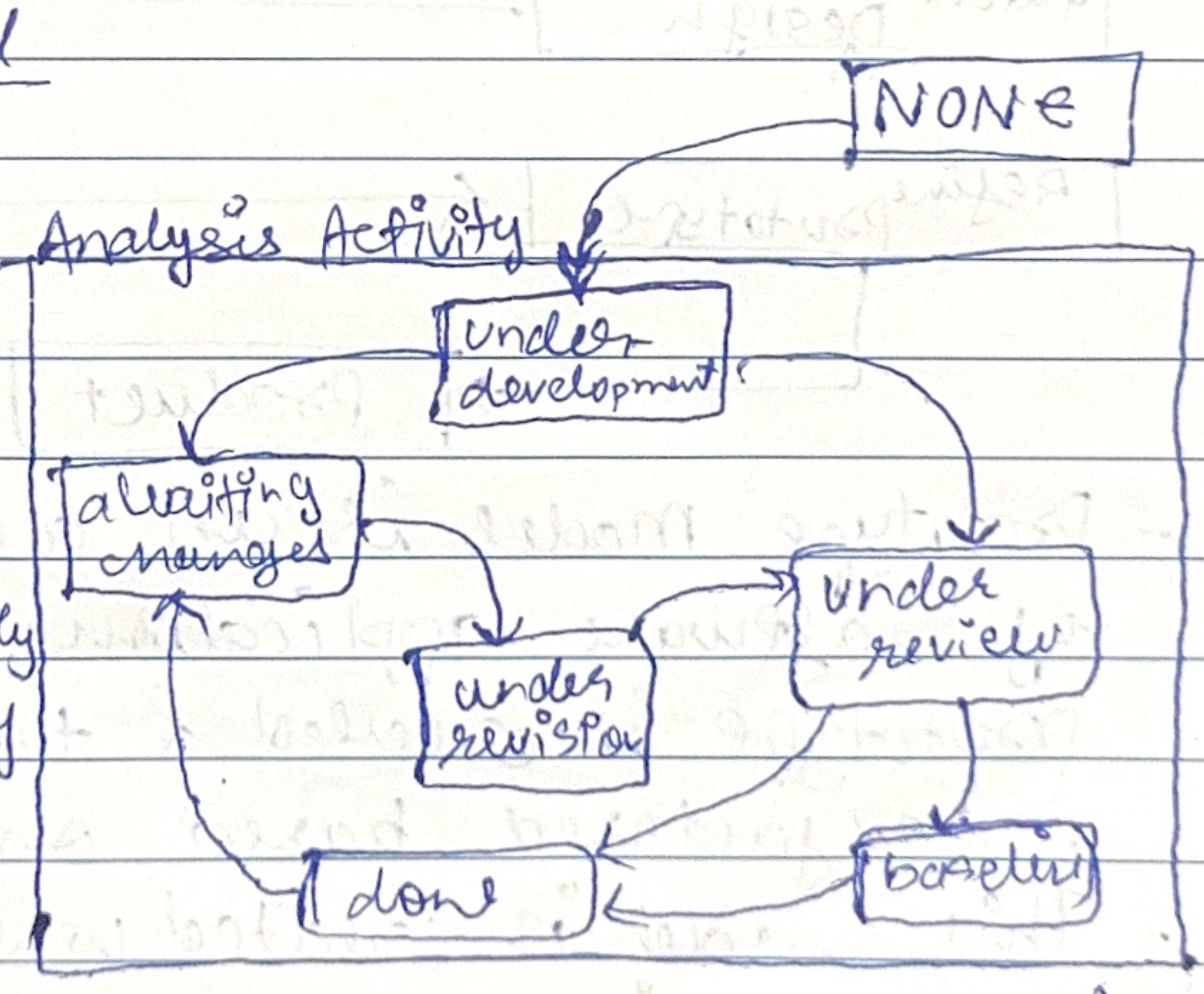
5. Incremental - model.



- Incremental model is divided into several increments and the same phases are followed in each increment. In simple language under this model a complex project is developed in many modules or builds.

Each subsequent release of a software module adds functions to the ~~pre~~ previous release. This process continues until the final software is obtained.

6. Concurrent model



- Concurrent model is where multiple development phases occur simultaneously overlapping instead of sequentially to speed up development & allow for continuous feedback, manage by defined states and triggers for activities.